Snapshot of October 29, 2026. This dossier describes the four repositories as they stood that day; the projects have moved on since, and the repositories are the authority. Underlined links lead to the sources, frozen at their state of that day.

DOSSIER 4 · THE SCENE

in.gnu6

An engine that decides what exists in a scene — positions, relations, time, constraints — and lets each display decide only *how* to show it.

RESEARCH | 2026.10.0-RESEARCH | NO OSS LICENCE

16
ratified milestones, MQ-00 to MQ-15, for the documented mandate

9,370
frames presented in 125 s, none discarded — one local session

40
real Wayland events exchanged between Rust and the TypeScript engine

0
stable public APIs, no package published on npm

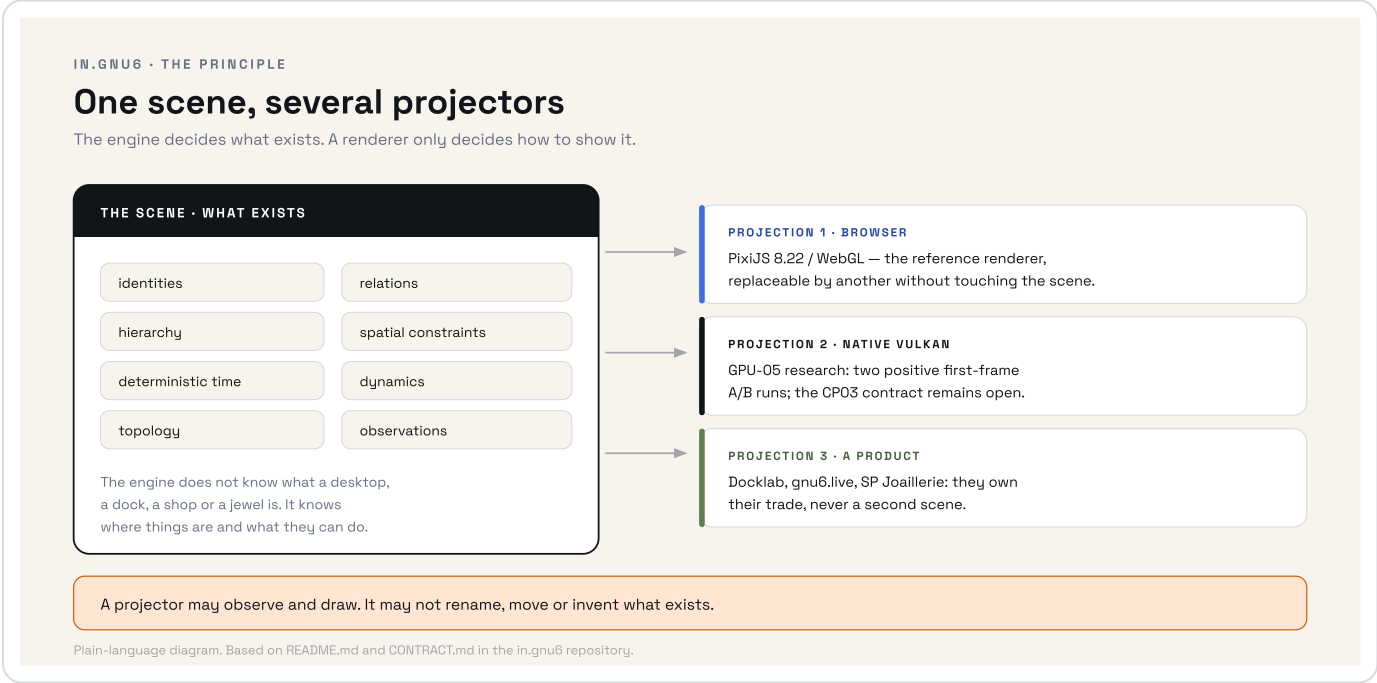
In one picture

THE ANALOGY

Think of a theatre. The **stage** knows where the actors, the sets and the passages are: it decides what exists and what is possible. The **projectors** light that stage with their own technique — one projects an image, another lights it at a grazing angle.

Changing projectors does not move the actors. That is the whole separation in.gnu6 stands for: the rendering is replaceable; the scene is not.

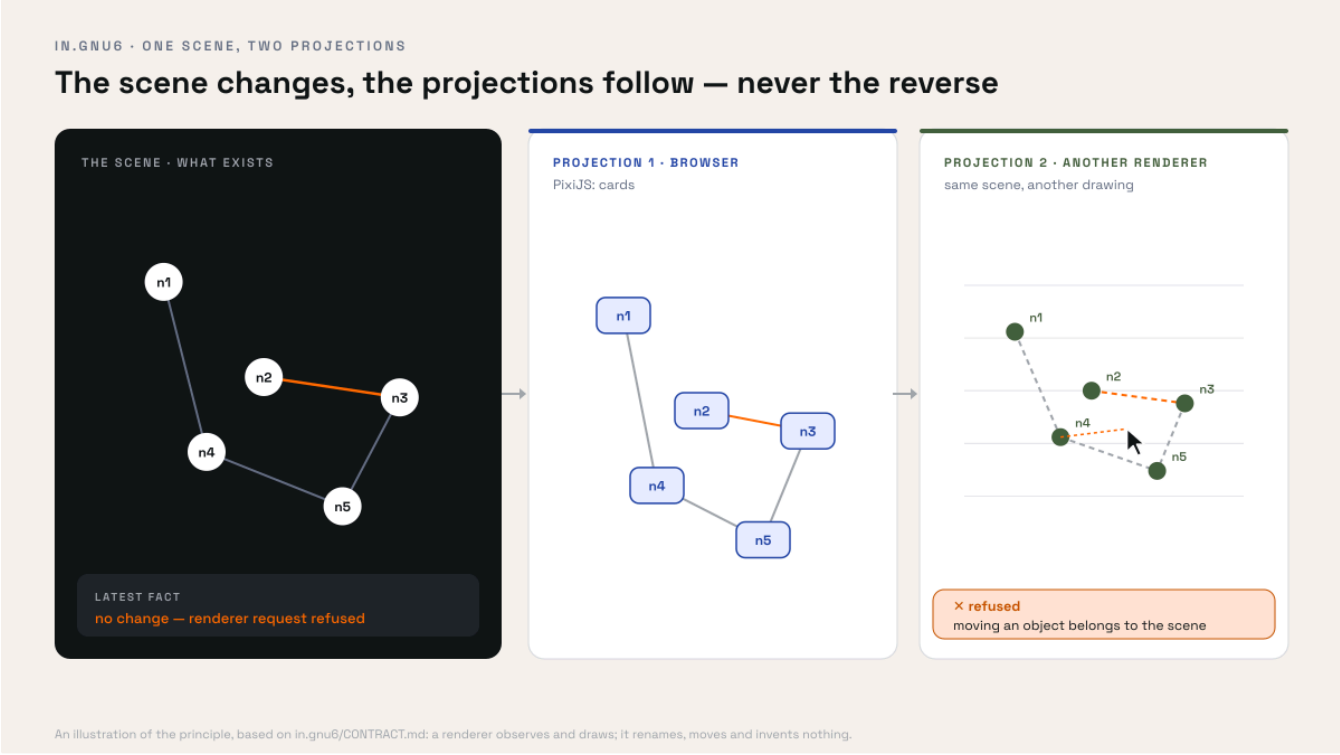
The principle



A single semantic authority. A renderer may observe and draw the scene; it may not rename, move or invent what exists. That is the central rule of the repository's contract.

Most graphics engines mix two questions: "what exists?" and "how do we draw it?". The mix holds until the day you want to change rendering engines, export a scene, or run two products on the same model — and then everything has to be rewritten.

in.gnu6 separates the two. A TypeScript core maintains the scene: identities, relations, hierarchy, spatial constraints, deterministic time, dynamics, topology, deformations, queries and reproducible observations. The renderers — a browser with PixiJS, a native Vulkan engine, or a consuming product — only project that state.



One way only. The scene changes an object or a relation: both projections follow, each with its own drawing. A projection that tries to move an object is refused — moving belongs to the scene. An illustration of the principle, based on `CONTRACT.md`.

What the engine owns — and what it ignores

It owns

- the identity of a scene and of its objects;
- the relations and hierarchy between them;
- spatial and admissibility constraints;
- deterministic time and dynamics;
- topology and its changes;
- reproducible observations.

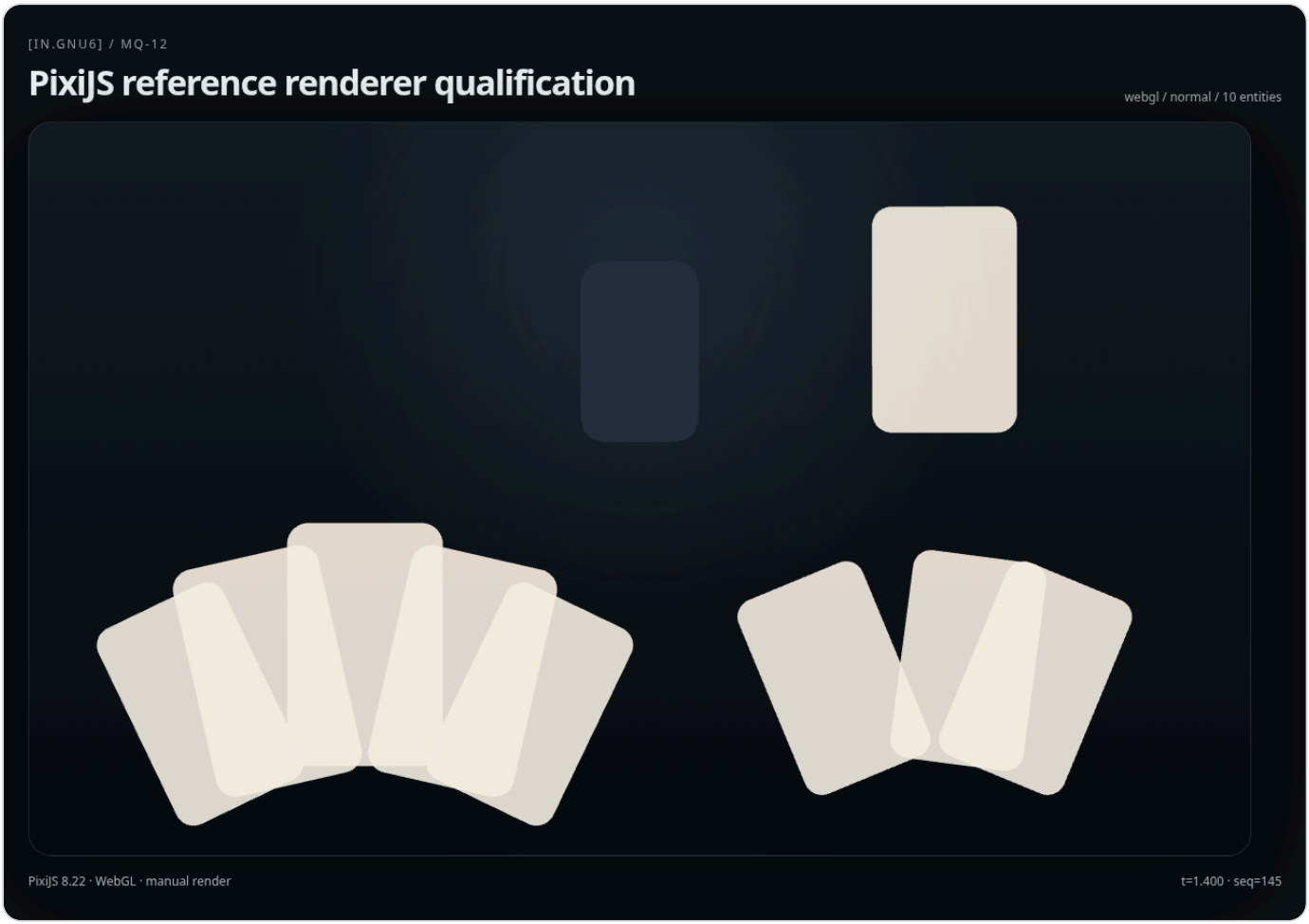
It ignores

- what a desktop, a dock or a workspace is;
- what a product, a shop or a basket is;
- what a piece of jewellery or a business action is;
- which commercial intent justifies a screen.

Those meanings belong to the products that consume the engine — not to the engine.

Three surfaces, three levels of truth

SURFACE	WHAT IT IS	EVIDENCE CURRENTLY ADMISSIBLE
TypeScript core	The display-free model: scene, topology, space, time, interaction	16 ratified milestones; full gate at 29 files / 119 tests, Pixi build and 2 browser tests
PixiJS / WebGL	The reference projection in a browser, replaceable	Qualified by milestones MQ-11 (contract) and MQ-12 (browser)
Native Vulkan	Research into fast native rendering, outside the browser	Partial local evidence; the overall acceptance contract CP03 remains **open**



The reference projection, on screen. Screenshot from milestone MQ-12: the PixiJS renderer draws a ten-entity scene in a browser. The image shows the drawing; it is the core's contract, not the image, that settles what exists. Source: `docs/evidence/assets/`.

The figures, and what they do not prove

Two observations recur in the repository's documents, and they are interesting precisely because they are bounded.

Endurance: 9,370 frames presented in 124.994 seconds, without a single discarded frame — about 75 frames per second on average, with a p95 of 13.478 ms. Measured locally, through the Wayland compositor's presentation timestamps.

Real exchange: a Rust program and the TypeScript semantic projector exchanged 40 real Wayland events over a private Unix socket, with an exact sequence match and 1,497 frames presented in 20 seconds, without loss.

THE LIMITS, PLAINLY

- **A single configuration.** These measurements come from one NVIDIA reference machine. They guarantee nothing on another card, or on several.
- **No input-to-photon.** The delay between a click and the displayed pixel is not measured; the runs that came close did not observe a real click.
- **An unexplained anomaly.** An earlier run showed a 413 ms stall that was never explained. It stays on the record.
- **CP03 remains open.** The overall acceptance contract — sixty frames per second sustained, independent latency, audit — is not closed.

What the licence does not say

GOOD TO KNOW BEFORE REUSING

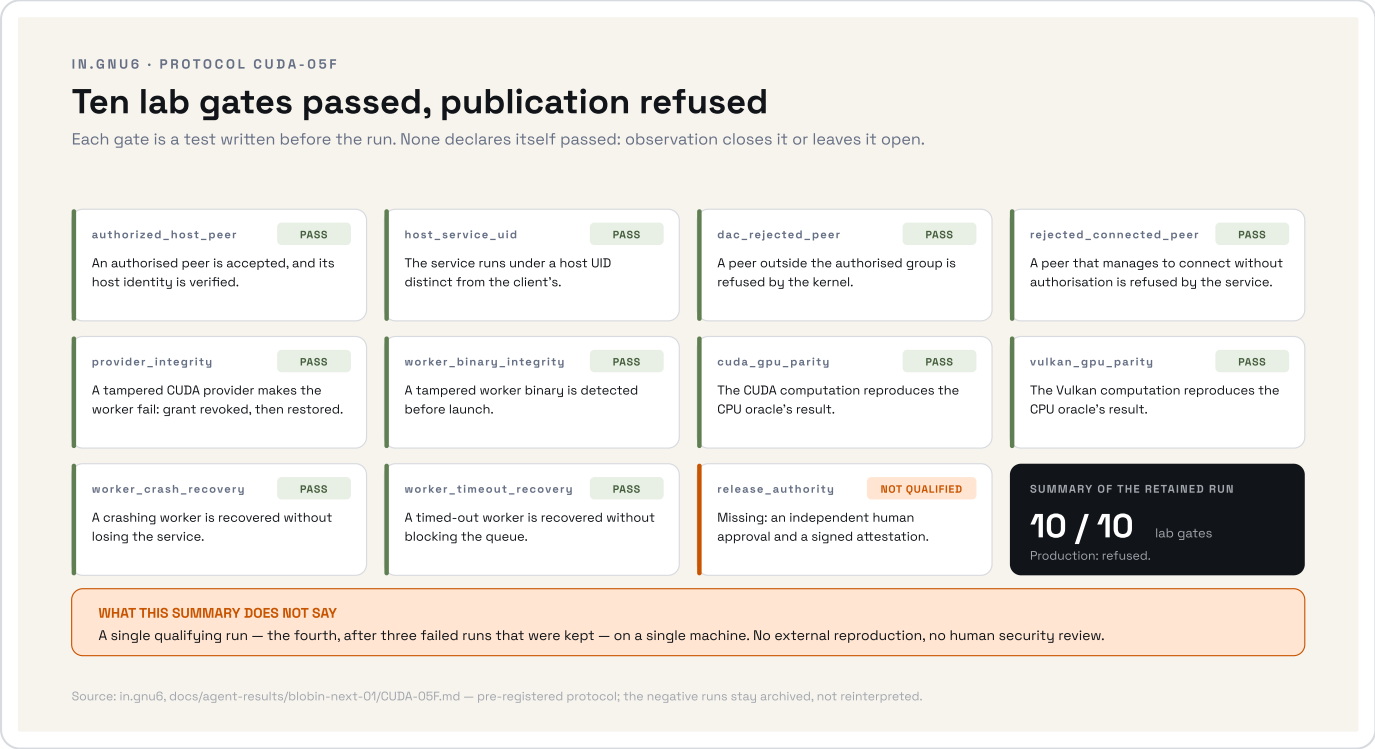
The repository has been public since October 10, 2026, but **its licence has not been decided**. The package file explicitly declares "UNLICENSED". A visible repository therefore does not grant a general right to copy or modify the code.

Two deliberate absences come on top: no stable public API, and nothing published on npm. Today the engine is to be assessed as a documented research programme, not as a dependency you install.

Why it is a programme of evidence

The repository does not present a list of features: it presents a list of **milestones to ratify**. Each must rest on a law, a falsification test, an execution observation — or remain explicitly open. A milestone can be rejected by a failing test, and the rejection stays written down.

A side experiment in the repository shows the method at work. A Rust supervisor hands a graphics computation to a separate process, under a distinct system account, and eleven gates are written *before* the run. Three acquisitions failed; they are kept. The fourth passes the ten lab gates — and the eleventh, the authorisation to publish, remains refused for lack of an independent human review.



Passing your tests does not authorise publishing. Protocol CUDA-05F: account isolation, binary integrity, GPU/CPU parity, recovery after a crash. Release authority is a separate gate: no automated test can pass it.

Two consuming products act as pressure: Docklab and a harness for a jewellery shop. Their role is not to validate the engine out of politeness, but to break it: if a product needs a second scene to work, the separation is wrong.

This discipline explains the figures at the top of the page: 16 ratified milestones do not mean "ready", and 9,370 frames without loss do not mean "smooth for you". Every number carries its frame.

What it changes, depending on who you are

Curious

You understand why an application can change its look without changing its content — and why "redoing the rendering" does not mean redoing the model.

Developer

A tested TypeScript core, a written renderer contract, a reference browser renderer, and a Vulkan experiment documented with its failures. Much of it can be checked without a graphics card.

Decision-maker or funder

The question is an industrial one: what does it cost on the day you must change rendering engines or plug in a second product? The answer here is an architecture, not a promise.

Going further

- TECHNICAL — THE REPOSITORY'S WORDS
- REPRODUCING WITHOUT AMPLIFYING AUTHORITY